What Causes Fed Rate Changes?

Is High/Low Unemployment The Biggest Factor Or Are There Other Causes?

Why This Matters.

Fed rates have a significant influence on the entire economy of the U.S and Americans are affected by it every day of the week, month and year. It affects how they are able to borrow money at what rate. This includes car payments, mortgages, personal loans and also investment decisions. While rate hikes and cuts influence the economy and its performance on a domestic level, it cannot entirely counter geopolitics that potentially cause the economy to turn sideways until they are resolved ,but can counter anything temporary such as economic policy changes within the government.

Regression Results

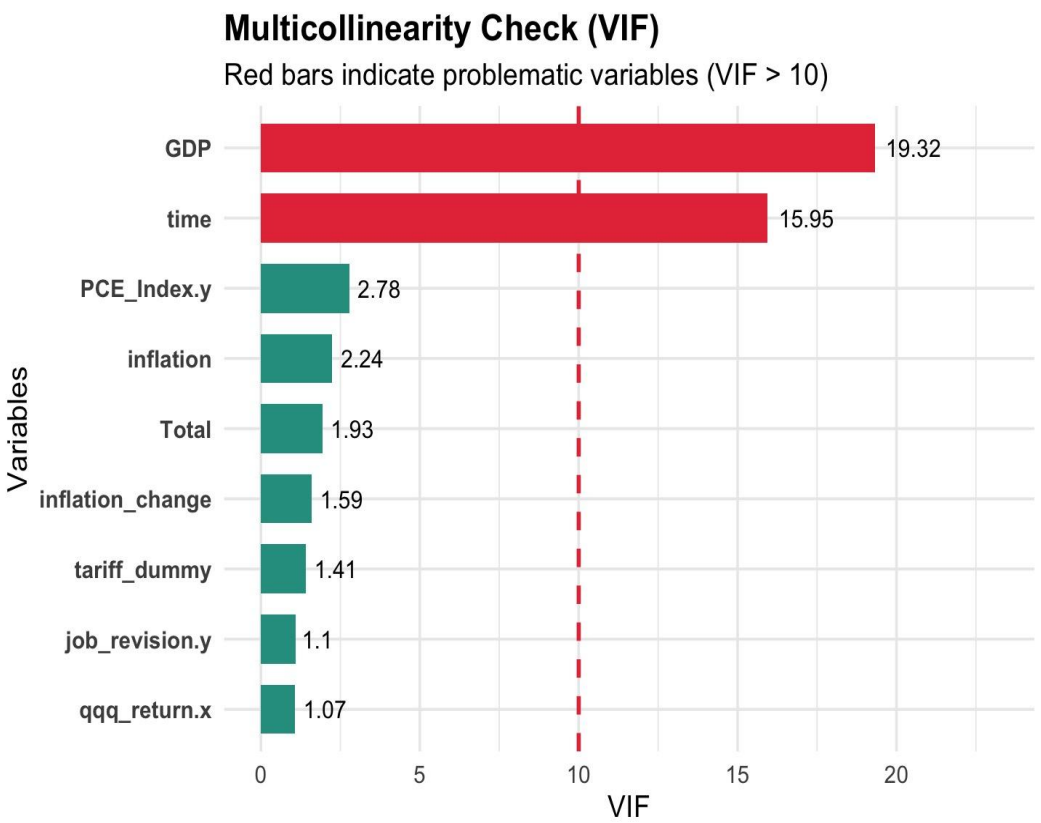
- Regression 7 has a very high R2, which indicates either there is a variable that explains the dependant variable or a problem exists .
- Unemployment remains one of the most significant factors across all models.

	Dependent variable:						
	fed_rate						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Total	-0.529***	-0.531***	-0.579***	-0.569***	-0.601***	-0.613***	-0.282***
	(0.049)	(0.051)	(0.053)	(0.060)	(0.060)	(0.060)	(0.037)
inflation		-0.294	-0.435	-0.520	-0.650	-0.698	-0.085
		(0.405)	(0.403)	(0.497)	(0.492)	(0.491)	(0.275)
inflation_change		0.038	0.119	0.174	0.293	0.273	0.040
		(0.409)	(0.404)	(0.439)	(0.431)	(0.430)	(0.240)
tariff_dummy			-0.988***	-0.948**	-0.554	-0.552	0.973***
			(0.369)	(0.385)	(0.384)	(0.382)	(0.224)
PCE_Index.y				0.036	0.206	0.228*	-0.311***
				(0.117)	(0.125)	(0.125)	(0.074)
job_revision.y					-0.001	-0.001*	-0.00005
					(0.001)	(0.001)	(0.0004)
time					-0.007***	-0.008***	-0.081***
					(0.002)	(0.002)	(0.004)
qqq_return.x						0.034*	0.029**
						(0.020)	(0.011)
GDP							0.001***
							(0.0001)
Constant	4.787***	4.833***	5.241***	5.116***	5.689***	5.742***	-10.278***
	(0.301)	(0.344)	(0.372)	(0.518)	(0.549)	(0.547)	(0.788)
Observations	240	236	236	235	228	228	228
R2	0.330	0.329	0.349	0.347	0.374	0.382	0.809
Adjusted R2	0.328	0.320	0.337	0.333	0.354	0.359	0.801
Residual Std. Error	1.631 (df = 238)	1.635 (df = 232)	1.614 (df = 231)	1.619 (df = 229)	1.573 (df = 220)	1.567 (df = 219)	0.874 (df = 218)
F Statistic	117.442*** (df = 1; 238)	37.843*** (df = 3; 232)	30.923*** (df = 4; 231)	24.389*** (df = 5; 229)	18.771*** (df = 7; 220)	16.918*** (df = 8; 219)	102.304*** (df = 9; 218)

Running Diagnostics

- Multicollinearity exists only in Model 7 when the “GDP” variable was in the regression alongside time
- In a trend context, since GDP is always in an upward trend and time also is in an upward trend, when comparing regressions R thinks that they are the same variable, but contextually they are not

Even trying to log the variable GDP to represent change still does not solve the multicollinearity issue but rather worsens it.



Total	inflation	inflation_change	tariff_dummy	PCE_Index.y
2.266748	2.246673	1.589018	1.405414	2.850069
job_revision.y	time	qqq_return.x	log_GDP	
1.098013	42.669811	1.070251	50.496658	

Running Diagnostics Multicollinearity

- It is best that we do not run regressions with time and GDP together
- Rather run them separately
- We see that GDP is actually a less important variable when running regressions separately with the same variables involved.
- Multicollinearity was no longer detected after this adjustment

	Dependent variable:	
	fed_rate	
	(1)	(2)
Total	-0.551***	-0.613***
	(0.065)	(0.060)
inflation	-0.375	-0.698
	(0.509)	(0.491)
inflation_change	0.063	0.273
	(0.444)	(0.430)
tariff_dummy	-0.852**	-0.552
	(0.388)	(0.382)
job_revision.y	-0.001	-0.001*
	(0.001)	(0.001)
GDP	0.00001	
	(0.00003)	
time		-0.008***
		(0.002)
qqq_return.x	0.020	0.034*
	(0.021)	(0.020)
PCE_Index.y	0.021	0.228*
	(0.134)	(0.125)
Constant	4.801***	5.742***
	(0.818)	(0.547)
Observations	228	228
R2	0.340	0.382
Adjusted R2	0.316	0.359
Residual Std. Error (df = 219)	1.619	1.567
F Statistic (df = 8; 219)	14.121***	16.918***

Total	inflation	inflation_change	tariff_dummy	job_revision.y	time	qqq_return.x	PCE_Index.y
1.608441	2.221771	1.587162	1.277128	1.080972	1.576267	1.070241	2.470507

Total	inflation	inflation_change	tariff_dummy	job_revision.y	GDP	qqq_return.x	PCE_Index.y
1.736380	2.240015	1.590233	1.236133	1.094274	1.909397	1.069651	2.669758

Logged Regressions

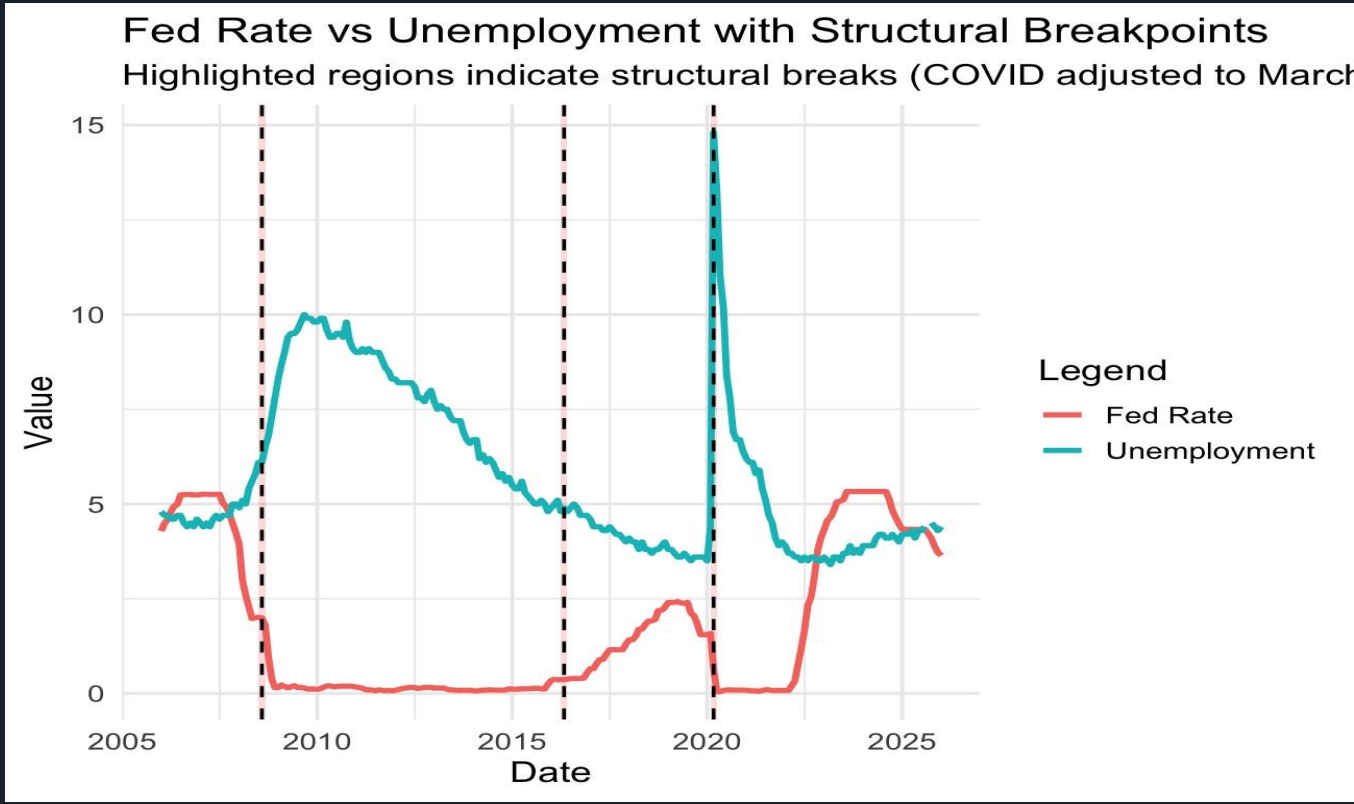
- Logged Regressions significantly improved how models performed.
- Models 10-13 (1-4) show how removing some logged. variables affected performance and understanding.
- No Multicollinearity detected.

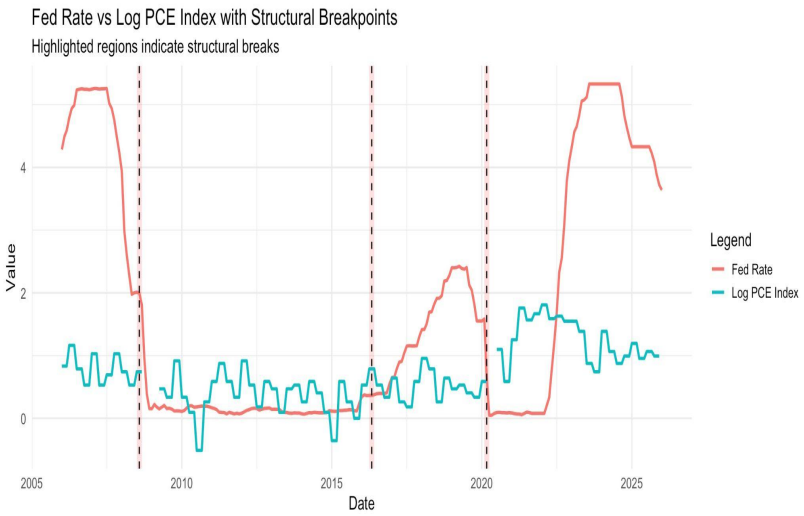
	Dependent variable:			
	fed_rate			
	(1)	(2)	(3)	(4)
log_Total	-4.550***	-4.739***	-4.727***	
	(0.379)	(0.361)	(0.357)	
inflation	-1.817***	-0.302	-0.256	0.515
	(0.542)	(0.374)	(0.322)	(0.423)
inflation_change	0.963**	0.091		
	(0.461)	(0.373)		
tariff_dummy	-0.785**	-1.161***	-1.154***	0.357
	(0.350)	(0.347)	(0.345)	(0.435)
job_revision.y	-0.001*	-0.001	-0.001	-0.0002
	(0.001)	(0.001)	(0.001)	(0.001)
time	-0.013***	-0.009***	-0.009***	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
qqq_return.x	0.029	0.033*	0.033*	-0.006
	(0.019)	(0.019)	(0.019)	(0.025)
log_PCE	1.315***			
	(0.298)			
Constant	10.363***	10.846***	10.817***	1.216***
	(0.813)	(0.762)	(0.752)	(0.267)
Observations	219	228	228	228
R2	0.504	0.453	0.453	0.019
Adjusted R2	0.485	0.436	0.438	-0.003
Residual Std. Error	1.419 (df = 210)	1.470 (df = 220)	1.467 (df = 221)	1.961 (df = 222)
F Statistic	26.674*** (df = 8; 210)	26.069*** (df = 7; 220)	30.534*** (df = 6; 221)	0.866 (df = 5; 222)

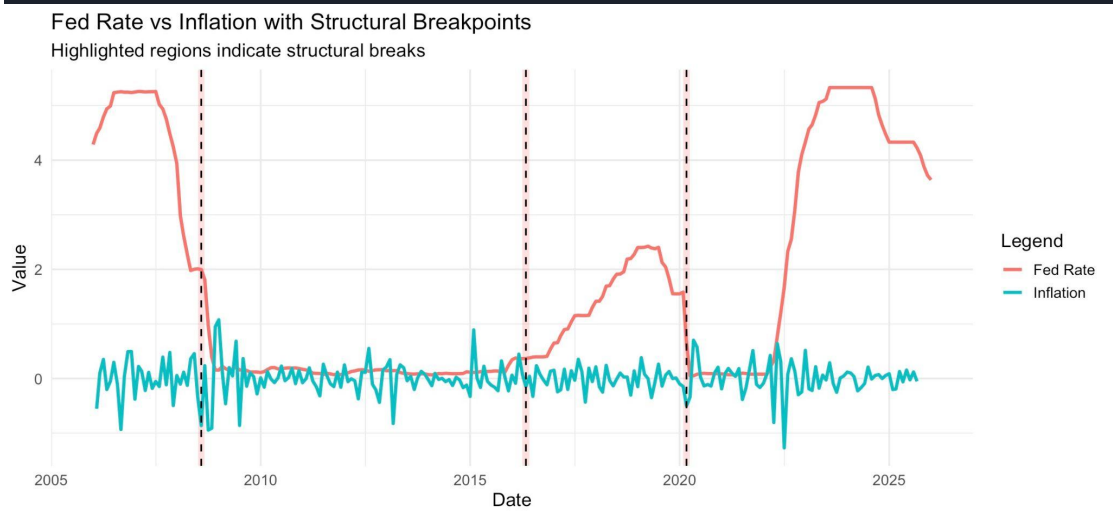
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log_Total      inflation inflation_change     tariff_dummy    job_revision.y             time     qqq_return.x
 1.774124       2.207666        1.701718         1.300463         1.034053         1.618808         1.041299
  log_PCE
 2.203511
log_Total      inflation inflation_change     tariff_dummy    job_revision.y             time     qqq_return.x
 1.653091       1.467816        1.355126         1.193961         1.057231         1.413617         1.056731
log_Total      inflation    tariff_dummy job_revision.y             time     qqq_return.x
 1.626217       1.092032         1.185015         1.047644         1.412714         1.051167
inflation    tariff_dummy job_revision.y             time     qqq_return.x
 1.056295         1.055320         1.037941         1.092945         1.024811
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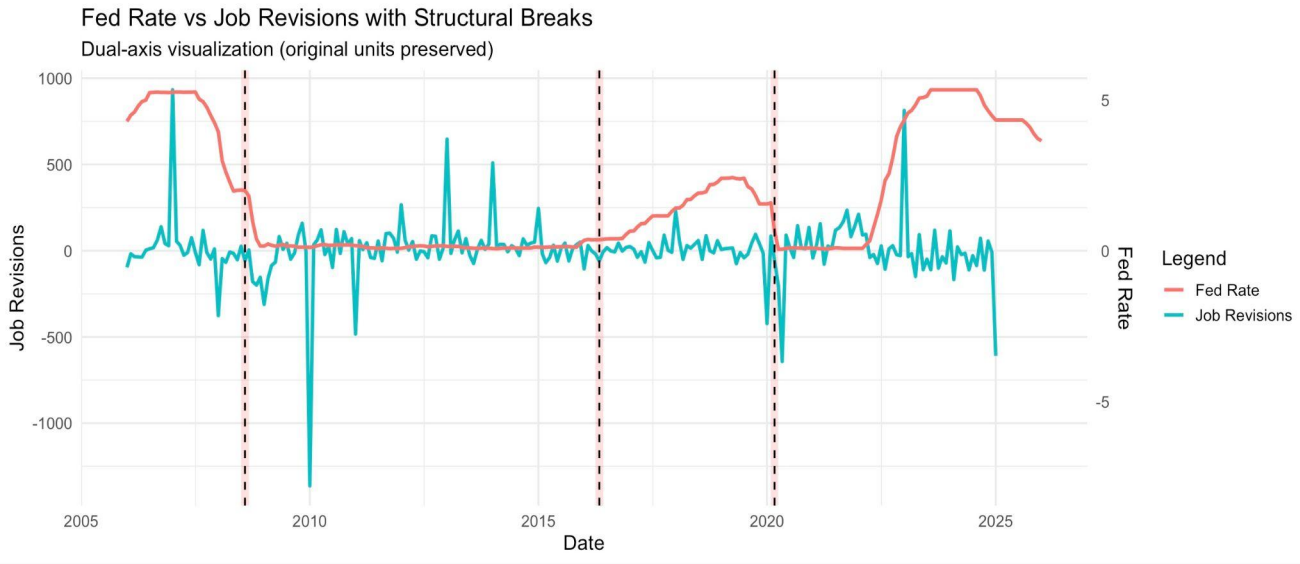
Running Diagnostics
Structural Breaks

- There are 3 structural breaks found.
- 2 Of them being the 2008 recession and 2020 COVID
- A Break at May 1st 2016.
- All Models when tested for structural breaks all had breaks detected as these variables are over a time series.




Fed Rate vs Log PCE Index with Structural Breakpoints
Highlighted regions indicate structural breaks
Value
Date
Legend
Fed Rate
Log PCE Index

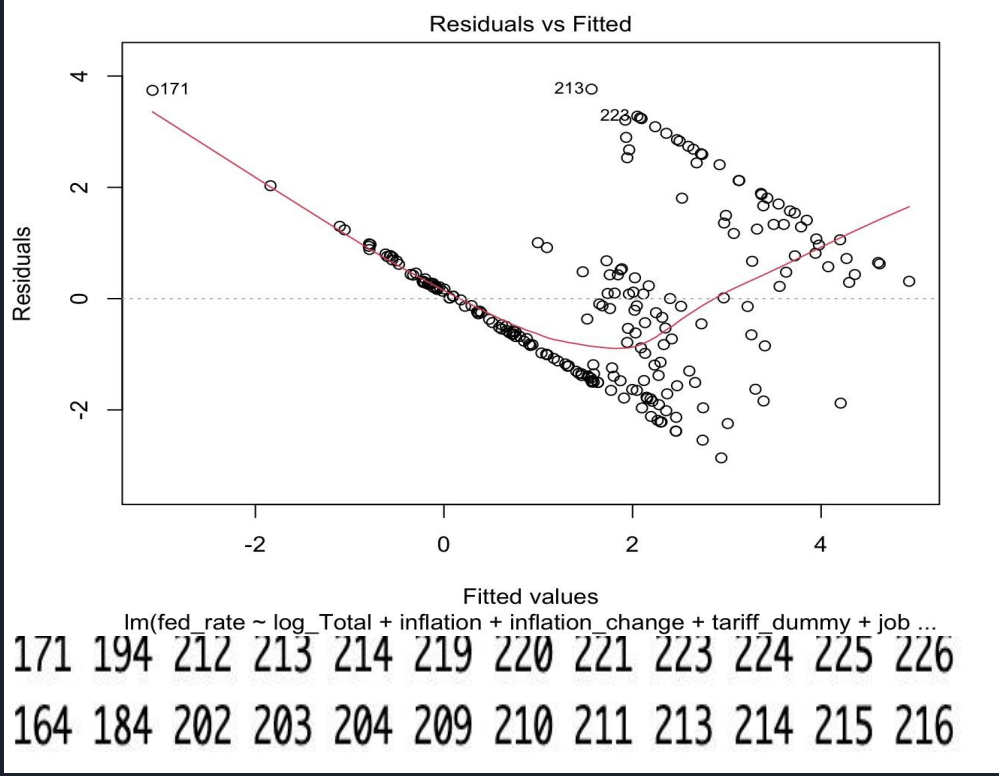

Fed Rate vs Inflation with Structural Breakpoints
Highlighted regions indicate structural breaks
Value
Date
Legend
Fed Rate
Inflation


Fed Rate vs Job Revisions with Structural Breaks
Dual-axis visualization (original units preserved)
Job Revisions
Fed Rate
Date
Legend
Fed Rate
Job Revisions

Outliers

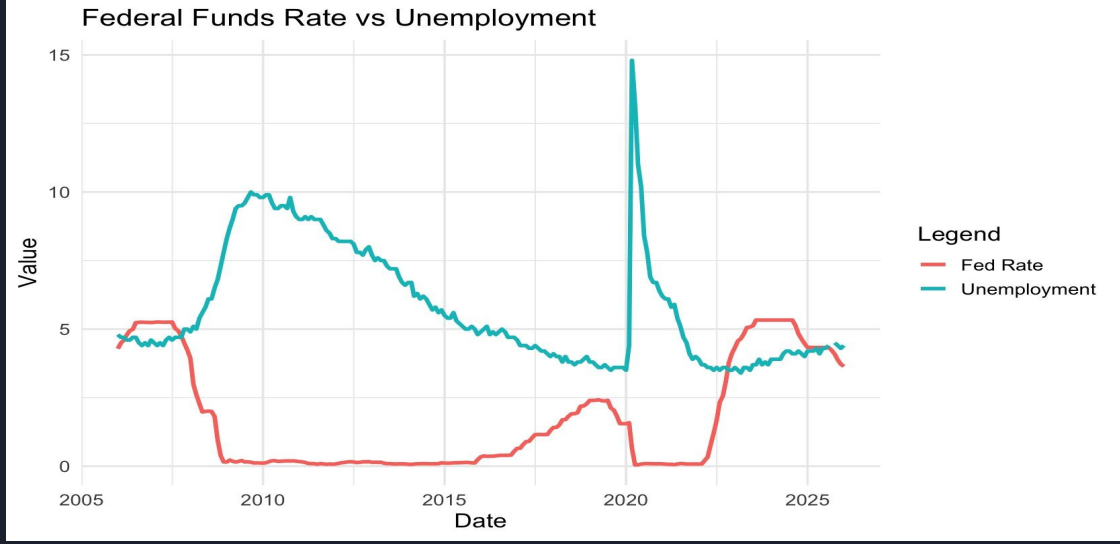
- Outliers existed in observations 171,194,213, in the logged regression.
- 171= start of pandemic
- 194 Major Inflation
- 213= slowing inflation

The cause of a lot of these outliers are from the variable QQQ returns where some months had massive jumps and dips and as well as inflation.



Key Findings

- Unemployment was the largest factor in fed rates across all tested models. According to the graph, if unemployment is up or if the job market is looking a little rough on a consistent pattern or rates get lowered, but if the economy does well with consistency meaning unemployment drops raise rates.
- Job Revisions when running regressions stated it was not as significant however looking at the pattern of the graphs, it seems to have some pattern when it comes to bigger crises.



Policy Implications

At the end of the day, whoever may be the fed chair or those that help decide whether to increase rates or to cute rates should look at job revisions, inflation, and most importantly unemployment rates. If there is a consistent rise in unemployment, lower job revisions, and rising inflation over a monthly time frame, dropping rates should be a consideration to ease the overall economy and vice versa.

However major geopolitics can affect the economy in a big way to the point that feds can't do anything about it but just watch how it is affected. Dropping rates significantly at once can do more harm than good if the economy is still functioning even if geopolitics is in the way.

Some limitations will be that not all observations are being tested. Over a 20 year span and 250 revisions, only 219 are actually being observed when accounting for all variables and accounting for change overtime for some variables.

There could also be some endogeneity as inflation can make the fed react and vice versa.

Geopolitical effects also is a limitation

Timing in when the fed reacts to patterns are also delayed which is another limitation.

